King Faisal University

Department of Computer Engineering

IoT Air Quality Monitoring System

Project Report

Embedded Systems

CE 323

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Abstract

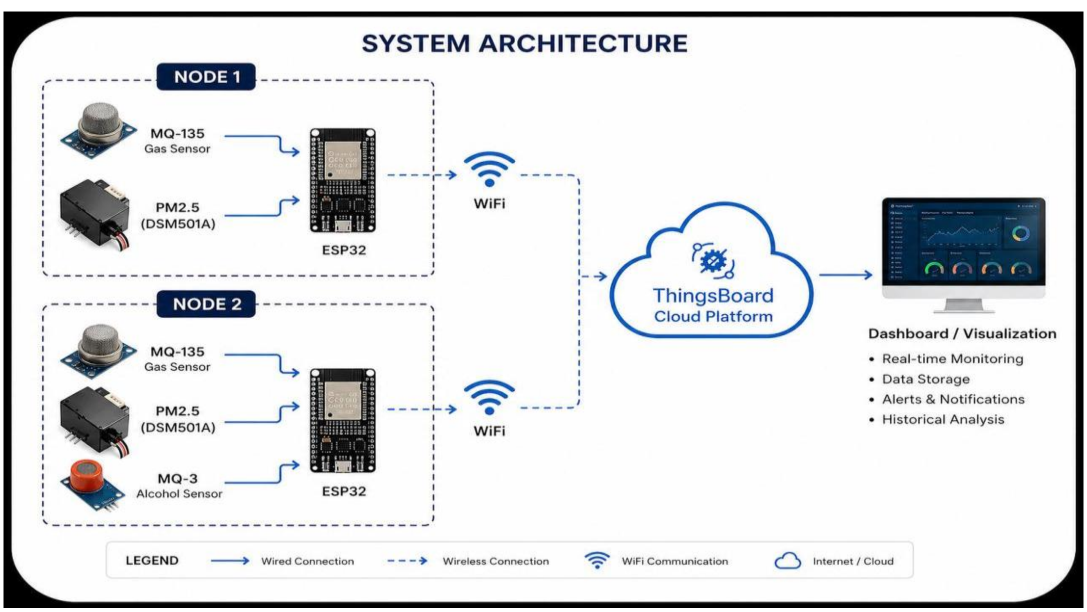
This project demonstrates the design of a system for measuring the air quality using ESP32 microcontroller along with MQ-135 and PM2.5 sensors. The system can detect the concentration of gases and PM2.5 dust in the air. The gathered data is sent through Wi-Fi to the Things Board cloud platform.

Problem Statement

Air pollution is dangerous in both an industrial setting and residential settings. Current methods of monitoring air pollution are quite costly and cannot be used in real time in a small-scale environment. The need, therefore, arises to design a system that can monitor air pollution continuously at relatively affordable costs. Such a system would measure the number of toxic elements in the atmosphere.

System Architecture

The system uses a distributed IoT-based architecture that includes two independent sensor nodes. First node is constructed using an ESP32 microcontroller that has been linked to an MQ-135 gas sensor and a PM2.5 particle matter sensor through the breadboard configuration. Second node is constructed using an ESP32 microcontroller that has been linked to an MQ-135 gas sensor, MQ-3 alcohol sensor and a PM2.5 particle matter sensor through the breadboard configuration. The ESP32 serves as the node for acquiring data and communicating. The data is wirelessly sent to the Things Board cloud platform through Wi-Fi connection. The cloud platform will aggregate the data received from the two nodes.



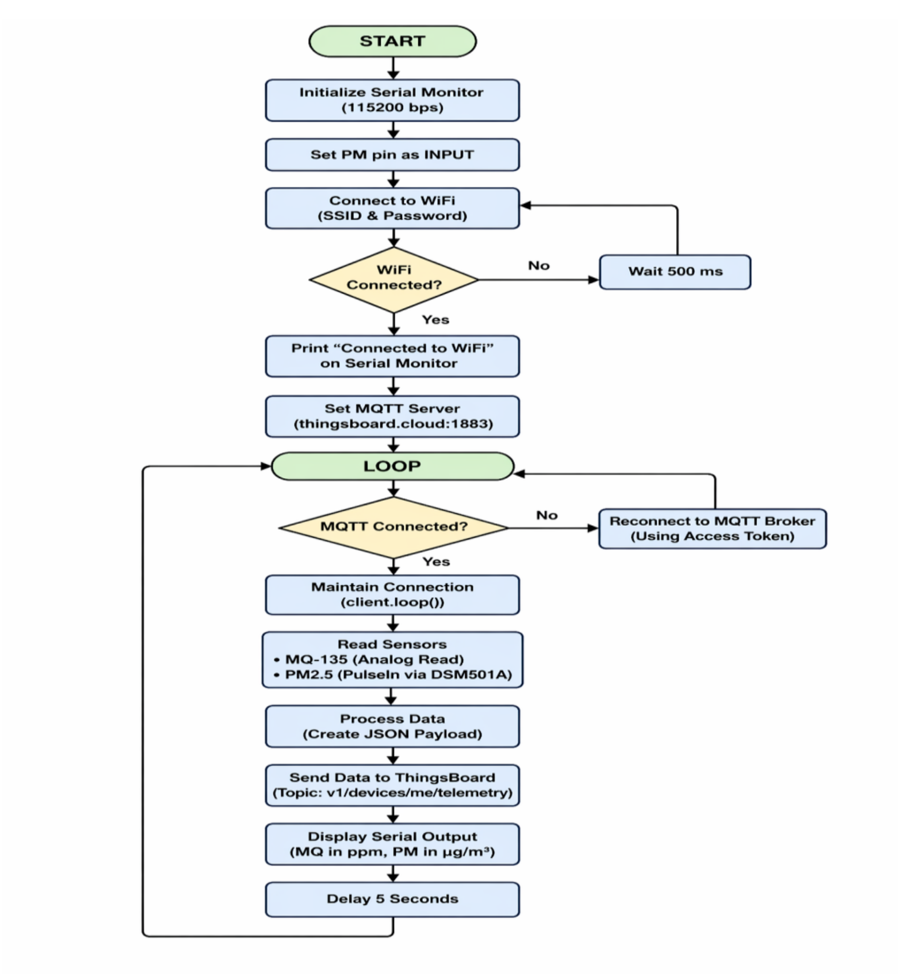
Hardware Design

The hardware design has two similar nodes using ESP32 microcontroller. Each node has an MQ-135 gas sensor that detects harmful gases and a PM2.5 sensor that determines the concentration of particulate matter. Node 2 has extra sensor MQ-3 alcohol sensor. The sensors are linked to ESP32 microcontroller via breadboard links. The ESP32 microcontroller is the data acquisition unit since it acquires data from sensors and transmits it via Wi-Fi.

Software Design

This software implementation has been done using ESP32 microcontroller on Arduino uno. This system keeps on receiving data from the MQ-135 gas sensor and PM2.5 particulate matter sensors. Once this data is received from these sensors, it is processed to make it in the JSON format. ESP32 connects to the Wi-Fi network and sends this processed data over the Things Board cloud platform using MQTT technology.

Flowchart



Algorithm

1. Start system
2. Initialize ESP32 and sensors
3. Connect to Wi-Fi
4. Set MQTT server

5. Loop continuously:
- Check MQTT connection
→ If not connected: reconnect
- Maintain connection (client.loop)
- Read sensors
- Process data
- Convert to JSON
- Publish via MQTT
- Wait 5 seconds

Results and Discussion

The system successfully transmits real-time air quality data to the Things Board dashboard. The Serial Monitor displays gas concentration in ppm and dust density in μg/m^3. The results show stable and consistent readings, indicating proper sensor functionality and reliable communication with the cloud platform.

Table 2.1 for samples of node 1 and Table 2.2 for node 2 as shown below

Table 2.1

Time/Sensor	MQ-135	PM2.5
8 PM	430	2.2
9 PM	479	6.8
10 PM	658	2.4

Table 2.2

Time/Sensor	MQ-135	PM2.5	MQ-3
8 PM	150	1600	170
9 PM	230	4300	270
10 PM	480	7800	520

Time series chart for MQ-135 in node1 as shown below in figure 3.1

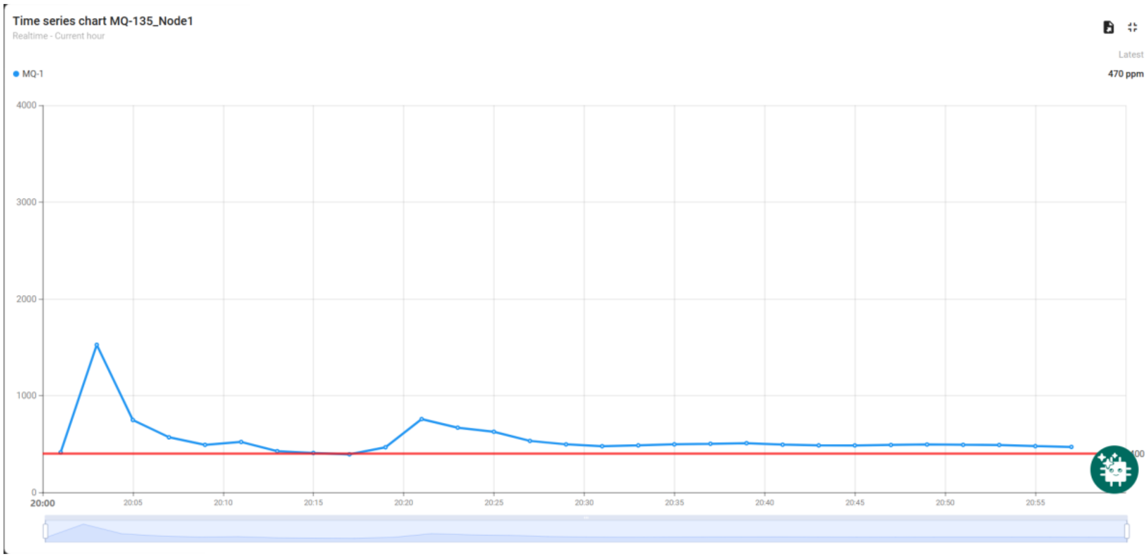


Figure 3.1

Time series chart for PM2.5 in node1 as shown below in figure 3.2

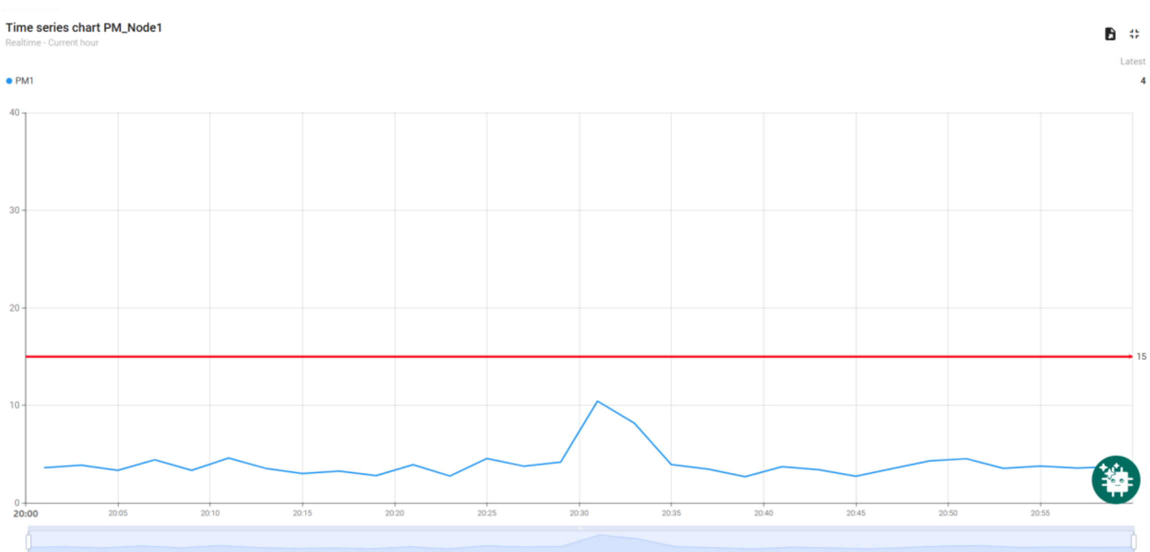


Figure 3.2

Time series chart for MQ-135 in node2 as shown below in figure 3.3

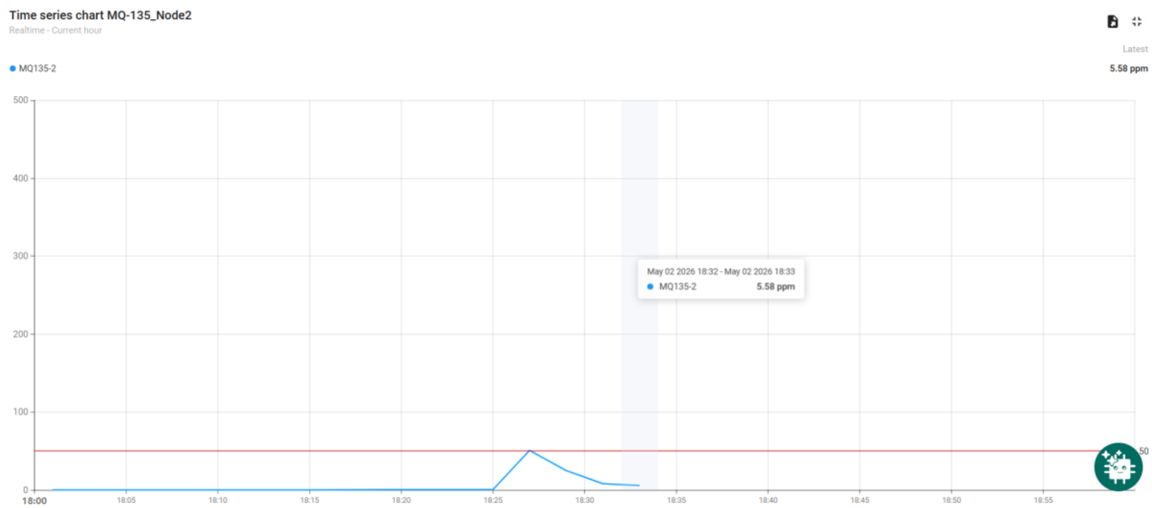


Figure 3.3

Time series chart for MQ-3 in node2 as shown below in figure 3.4

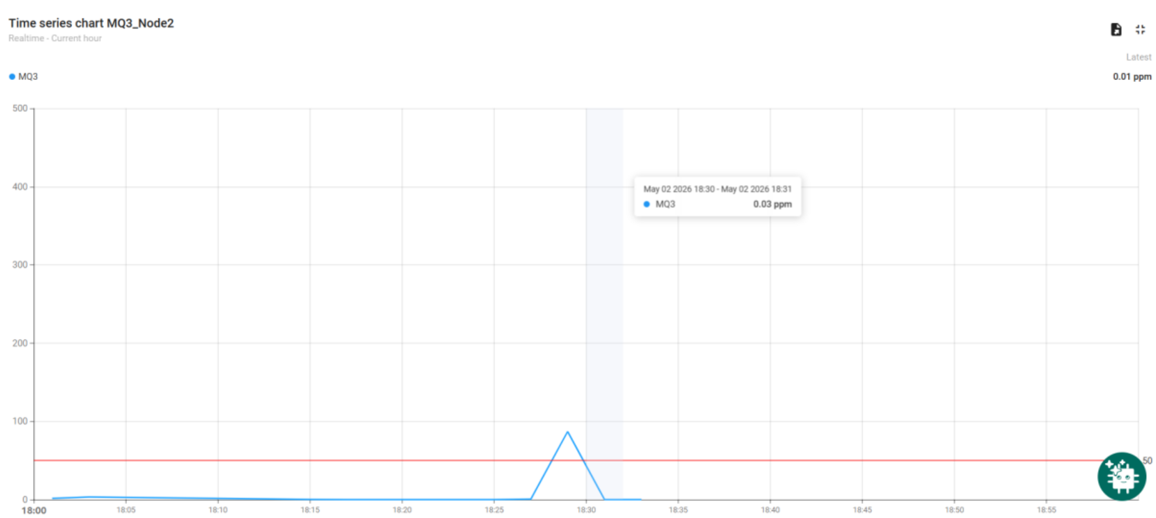


Figure 3.4

Time series chart for PM2.5 in node1 as shown below in figure 3.5

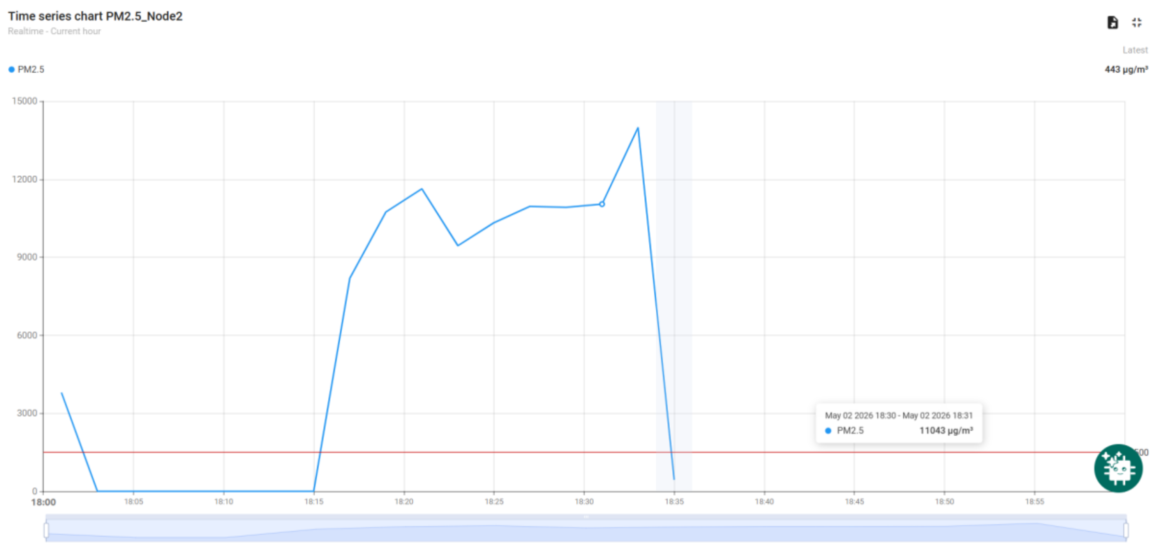


Figure 3.5

Conclusion

The project effectively deployed an air quality monitoring system based on ESP32-powered sensor nodes. In the system, the environmental data is collected by distributed nodes with embedded MQ-135 gas sensors, MQ-3 alcohol sensor and PM2.5 sensors. Then, the gathered data is uploaded to the Things Board cloud-based dashboard using Wi-Fi connectivity.

Overall, the air quality monitoring system operates reliably in real-time monitoring of environmental parameters. Also, the proposed system serves as an affordable alternative for measuring air quality at both residential and industrial sites. Some potential enhancements for future work could involve implementing predictive models through machine learning methods and advanced techniques for improving sensor accuracy.

GitHub link:

https://github.com/Eng-Mahdi47/Embedded-Systems.git